Q1.

The kinetic energy, E, of a compound pendulum is given by

$$E = \frac{1}{2}I\omega^2$$

where ω is the angular speed and I is a quantity called the moment of inertia.

(a) Show that for this formula to be dimensionally consistent then I must have dimensions ML^2, where M represents mass and L represents length.

(2)

(b) The time, T, taken for one complete swing of a pendulum is thought to depend on its moment of inertia, I , its weight, W, and the distance, h, of the centre of mass of the pendulum from the point of suspension.

The formula being proposed is

$$T = kI^{\alpha} W^{\beta} h^{\gamma}$$

where k is a dimensionless constant.

Determine the values of α, β and γ.

(3)
(Total 5 marks)

Q2.

A tank full of liquid has a hole made in its base.

Two students, Sarah and David, propose two different models for the speed, v, at which liquid exits the tank.

David thinks that v will depend on the height of the liquid in the tank, h, the acceleration due to gravity, g, and the density of the liquid, ρ, such that $v \propto g^a h^b \rho^c$ where a, b and c are constants.

Sarah thinks that v will not depend on the density of the liquid and suggests the model $v \propto g^a h^b$

(a) By considering dimensions, explain which student's model should be rejected.

(2)

(b) Find the values of the constants in order for the model that you did **not** reject in part **(a)** to be dimensionally consistent.

(2)
(Total 4 marks)

Q3.

The time, t, for a single vibration of a piece of taut string is believed to depend on

the length of the taut string, l,
the tension in the string, F,
the mass per unit length of the string, q, and
a dimensionless constant, k,

such that

$$t = kl^{\alpha} F^{\beta} q^{\gamma}$$

where α, β and γ are constants.

By using dimensional analysis, find the values of α, β and γ.

(Total 5 marks)

Q4.

A ball of mass m is travelling vertically downwards with speed u when it hits a horizontal floor. The ball bounces vertically upwards to a height h.

It is thought that h depends on m, u, the acceleration due to gravity g, and a dimensionless constant k, such that

$$h = km^{\alpha}u^{\beta}g^{\gamma}$$

where α, β and γ are constants.

By using dimensional analysis, find the values of α, β and γ.

(Total 5 marks)

Q5.

The speed, v m s^{-1}, of a wave travelling along the surface of a sea is believed to depend on

the depth of the sea, d m,
the density of the water, ρ kg m^{-3},
the acceleration due to gravity, g, and
a dimensionless constant, k

so that

$$v = kd^{\alpha}\rho^{\beta}g^{\gamma}$$

where α, β and γ are constants.

By using dimensional analysis, show that $\beta = 0$ and find the values of α and γ.

(Total 6 marks)

Q6.

The magnitude of the gravitational force, F, between two planets of masses m_1 and m_2 with centres at a distance x apart is given by

$$F = \frac{Gm_1m_2}{x^2}$$

where G is a constant.

(a) By using dimensional analysis, find the dimensions of G.

(3)

(b) The lifetime, t, of a planet is thought to depend on its mass, m, its initial radius, R, the constant G and a dimensionless constant, k, so that

$$t = km^{\alpha}R^{\beta}G^{\gamma}$$

where α, β and γ are constants.

Find the values of α, β and γ.

(5)
(Total 8 marks)

Q7.

The time T taken for a simple pendulum to make a single small oscillation is thought to depend only on its length l, its mass m and the acceleration due to gravity g.

By using dimensional analysis:

(a) show that T does **not** depend on m;

(3)

(b) express T in terms of l, g and k, where k is a dimensionless constant.

(4)
(Total 7 marks)

Q8.

An expression for calculating the volume per second of liquid flowing through a cylindrical pipe is of the form

$$kl^{-1}r^{a}p^{b}\eta^{c}$$

where k is a dimensionless constant;
l metres is the length of the pipe;
r metres is the radius of the pipe;
p is the excess pressure, measured in N m^{-2};
and η is the viscosity of the liquid measured in kg $m^{-1}s^{-1}$.

(a) (i) Write down the dimensions of η.

(1)

(ii) Determine the dimensions of p.

(2)

(b) By using dimensional analysis, find the values of a, b and c.

(6)
(Total 9 marks)

Q9.

The gravitational force exerted between two spheres of masses m_1 and m_2 is

$$\frac{Gm_1m_2}{r^2}$$

where G is a constant and r is the distance between the centres of the two spheres.

(a) Find the dimensions of G.

(3)

(b) The speed, v, of a satellite orbiting a planet of mass m is given by

$$v = \frac{G^\alpha m^\beta}{r^\gamma}$$

where r is the radius of the orbit and α, β and γ are constants. Find the values of α, β and γ for this equation to be dimensionally consistent.

(4)
(Total 7 marks)